



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (Even Semester)

Degree, Semester & Branch : B.Tech,IV & AI & DS
Course Code & Title : CS3591 & Computer Networks
Name of the Faculty member : Mrs. C.Usharani, AP/AD

Active learning practices: JIGSAW Cooperative Learning Technique

Theme of Discussion: User Datagram Protocol(UDP)

Date & Time: 10.03.2025 & 12.20 pm to 1.05 pm.

➤ **Justification:**

The **User Datagram Protocol (UDP)** is a key transport layer protocol in computer networks, providing a lightweight, connectionless communication method. It is widely used in scenarios where speed, low latency, and efficiency are prioritized over reliability and error correction. UDP has minimal protocol overhead compared to **Transmission Control Protocol (TCP)** because it does not include mechanisms for error correction, congestion control, or acknowledgment. UDP is well-suited for **real-time applications** such as VoIP (Voice over IP), live video streaming, and online gaming. The prime objective of Jigsaw is to encourage both self- and peer teaching, which requires students to understand the concept very clearly and engage in discussion and learning.

➤ **Time Allotted for the Activity:** 45 Minutes

➤ **Details of the Implementation:**

The students are given a group activity to discuss about the services or functions provided by UDP. The students are made to work into groups to discuss and clearly understand the User Datagram Protocol

➤ **Learning Outcomes:**

The students are able to Understand the Fundamentals of UDP, Analyze the Structure of UDP, Identify UDP Applications and Use Cases and Understand UDP's Connectionless Nature.

To improve listening, communication, and presentation skills.

➤ **Procedure:**

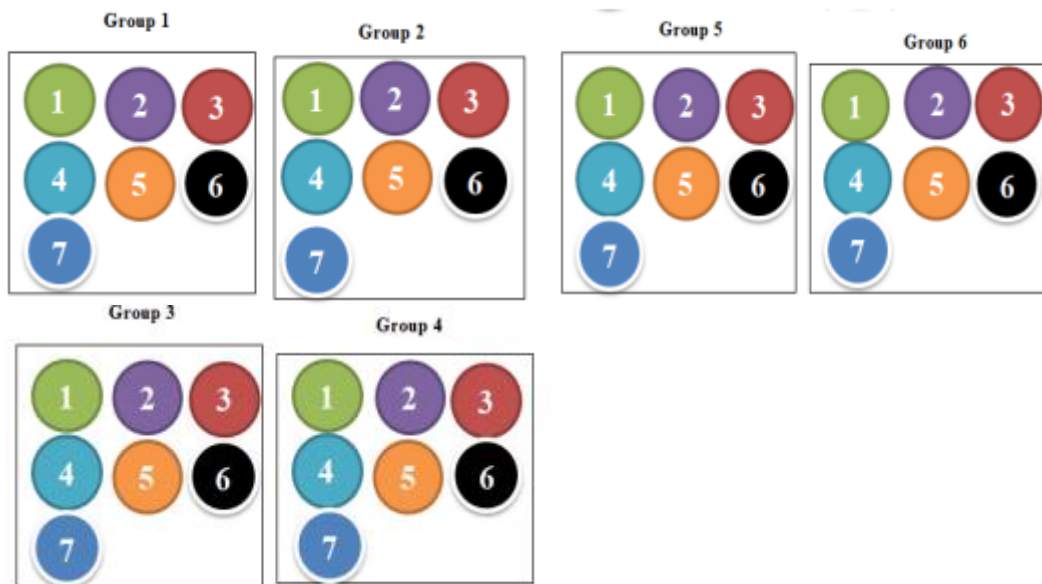
Step 1 (2 minutes)

- Students were divided into 6 groups with 7 members in each group based on the student's academic performance

Step 2 (2 minutes)

- Appoint one student from each group as the **leader**.
- Each student in a group is given a number from 1 to 6.

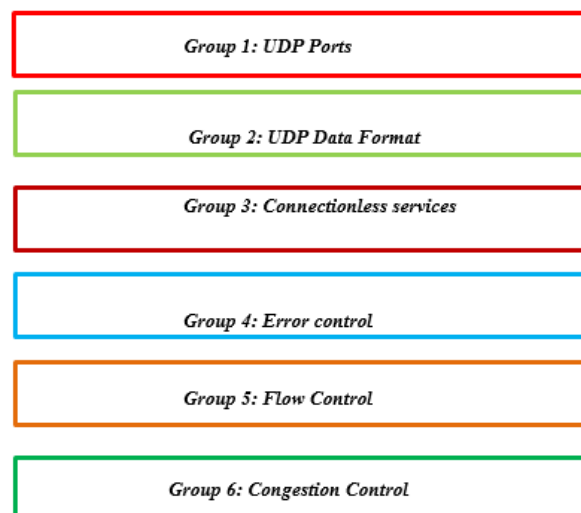
Jigsaw Groups



Step 3 (3 minutes)

- Divide students into 7-person Jigsaw groups as an expert group
- Divide the topic into 6 segments and assign it to each Expert Group.

The topics are given below:



Step 4 (15 minutes)

- Assign each student to learn one segment and time is allotted for discussion
- Give students time to read over their segment at least twice and become familiar with it.

Step 5 (2 minutes)

- Form temporary “expert groups” by having one student from each jigsaw group join other students assigned to the same segment.

Step 6 (15 minutes)

- Bring the students back into their jigsaw groups.
- Ask each student to explain her or his segment to the other team members in the group.
- Encourage others in the group to ask questions for clarification.

Step 7 (6 minutes)

- At the end of the session, the team leader summarizes the topic

Images / Screenshot of the practice:





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science
Academic Year 2024 – 2025 (Even Semester)

FEEDBACK Active Learning Best practices: JIGSAW Cooperative Learning Technique

Degree, Semester & Branch : B.Tech, IV & AI / DS
Course Code & Title : CS3591 & Computer Networks
Name of the Faculty member : Mrs.C.Usharani, AP/AD

Theme of discussion : User Datagram Protocol(UDP)
Date and Time : 10.03.2025 & 12.20 p.m to 1.05 p.m
Feedback collected in class and also through online

FEEDBACK QUESTIONS

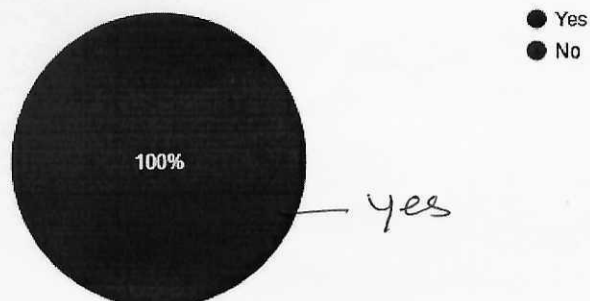
1. Does it encourage cooperative Learning Practices among yourself?
Yes / No
2. Connection with course material in a creative and engaging way to describe the
Services or functions of UDP.
Excellent /Good /Fair
3. Does this activity learning improve listening, communication and problem
solving skills?
Excellent /Good /Fair

Google form Link: <https://tinyurl.com/u39u44wt>

Feedback Analysis:

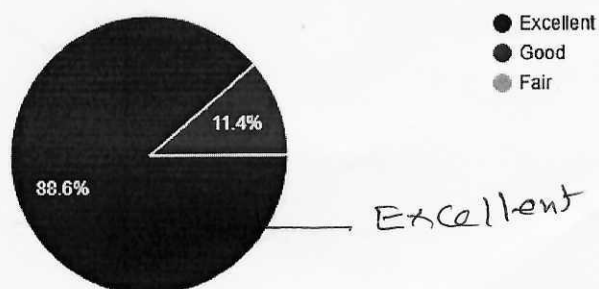
1.Does it encourage cooperative Learning Practices among yourself?

44 responses



2.Connection with course material in a creative and engaging way to describe the purpose of DNS.

44 responses



3.Does this activity learning improve listening, communication and problem solving skills?

44 responses



C. Vally
14/3/25
Faculty-Incharge

Wair
14/3
HOD/AD